EV BATTERY HEALTH PREDICTION

Mini Project Report | BCA Sem-III | Course 303-05

Student name: _____

Roll number: _____

Faculty guide: _____

Academic year: _____

Certificate

This report describes a beginner machine-learning project to predict battery State of Health (SoH, %). Complete the student details and get faculty approval before submission. The supplied Kaggle data is synthetic, not real battery-test data.

Student signature: _____ Date: _____

Faculty signature: _____ Date: _____

PROBLEM, OBJECTIVES, AND DATASET

Problem statement

Estimate battery State of Health (SoH, %) from vehicle age, charging cycles, temperature, fast-charge ratio, and discharge rate.

Objectives

1. Inspect and clean the supplied CSV.
2. Explore data with six charts.
3. Compare Linear, Ridge (L2), and Lasso (L1) regression.
4. Evaluate with five-fold CV and held-out test data.
5. Save the best model and predict one sample.

Dataset: 10,000 rows and 13 columns; 8 number columns and 5 text columns.

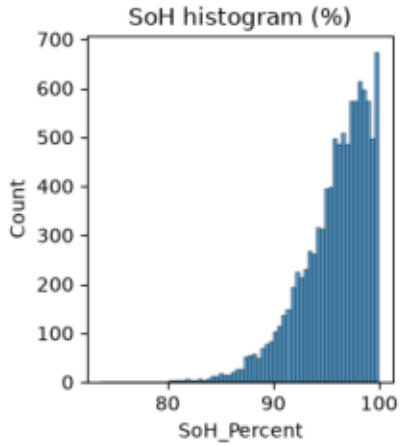
Units: capacity (kWh); age (months); cycles (cycles); temperature (Celsius); fast-charge ratio (unitless, 0 to 1); discharge (C-rate); resistance (ohms); SoH (%). Vehicle ID has no physical unit. Car model, battery type, driving style, and battery status are categories.

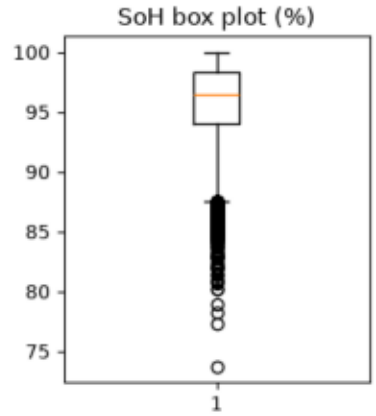
Important course note: the course asks for real-world data. The Kaggle author describes this dataset as synthetic simulations. Ask the teacher to approve it or use an approved measured dataset.

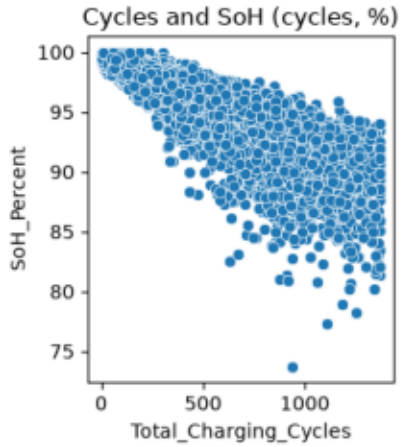
DATA PREPROCESSING AND EXPLORATORY DATA ANALYSIS

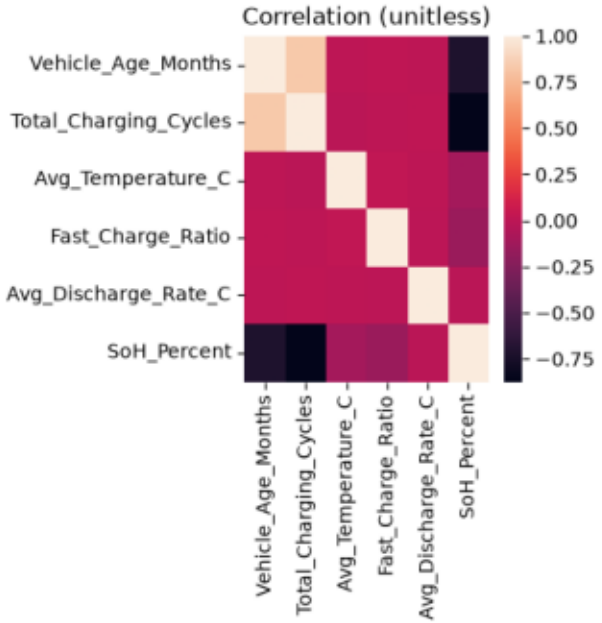
Missing values: 0 | Duplicate rows: 0 | IQR input values capped: 101 | Rows kept: 10,000

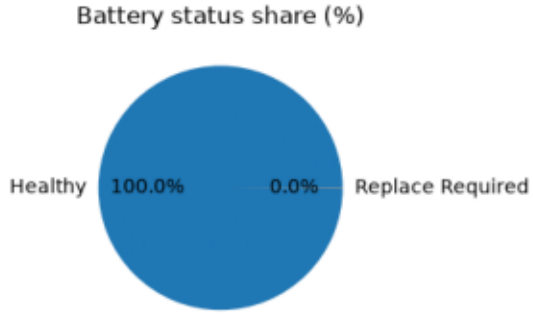
Five input measurements were clipped at IQR limits. Target SoH (%) was not changed.

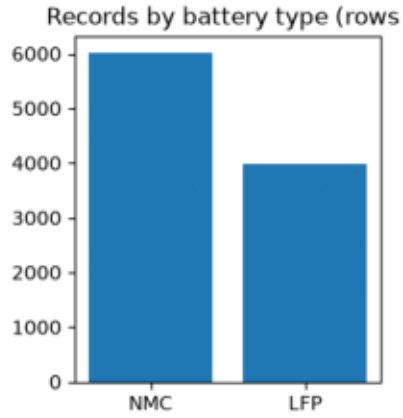












MODEL BUILDING, RESULTS, AND CONCLUSION

Inputs: age (months), cycles (cycles), temperature (Celsius), fast-charge ratio (unitless), discharge rate (C-rate). Target: SoH (%). Split: 8,000 training rows and 2,000 test rows. Model selection uses five-fold CV R-squared.

Model	CV R2	MAE %	MSE %2	RMSE %	Test R2
Linear	0.8008	1.0687	2.2163	1.4887	0.7916
Ridge	0.8008	1.0687	2.2163	1.4887	0.7916
Lasso	0.7894	1.0779	2.3229	1.5241	0.7815

CV R2 and test R2 are unitless. MAE and RMSE are in SoH percentage points (%); MSE is in % squared. Linear and Ridge tie at 4 decimals; Linear is selected.

One test sample: age 68 months, 432 cycles, 24.3 Celsius, fast-charge ratio 0.91, discharge rate 0.61 C-rate. Actual SoH: 97.31%. Predicted SoH: 95.41%.

Conclusion: the project demonstrates data inspection, cleaning, charts, regression, cross-validation, evaluation, and saving a model. Scores apply only to synthetic data.

Future scope: get teacher approval, use measured battery data, test on different batteries, and check whether predictions match real health measurements.

Dataset source: Kaggle, Electric Vehicle (EV) Battery Degradation and Charge, bertnardomariouskono.